

SQL DATABASE FUNDAMENTALS (345)

NATIONAL – 2019

TOTAL POINTS _____ ***(500 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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**Identify the letter of the choice that best completes the statement or answers the question.
Mark A if the statement is True. Mark B if the statement is False.**

1. Which of the following will produce the *average* of all salaries among employees?
 - a. Select avg(salary) from Employees_T
 - b. Select salary from avg(Employees_T)
 - c. Select sum(salary) from Employees_T where salary = avg
 - d. Select average salary from Employees_T
2. What are the commands that are used in order to insert data from one table to another?
 - a. INSERT...EXECUTE
 - b. INSERT INTO...ALTER
 - c. INSERT...SELECT
 - d. INSERT INTO...SELECT
3. The following syntax issues a call to the stored procedure *sp_indexinfo*
 - a. GO *sp_indexinfo*
 - b. CALL *sp_indexinfo*
 - c. EXEC *sp_indexinfo*
 - d. RUN *sp_indexinfo*
4. Which of the following are methods to backup a database?
 - a. Virtual Full
 - b. Incremental
 - c. Differential
 - d. All of the above
5. A unique index allows the ability to include duplicate values:
 - a. True
 - b. False
6. Besides tuples, which of the following objects are needed in order to make up a relationship?
 - a. Data Types
 - b. Primary Key
 - c. Attributes
 - d. None of the above
7. What organization developed SQL back in the late 1970s?
 - a. IBM
 - b. Microsoft
 - c. Oracle
 - d. Intel

8. Which of the following is *not* a relational database?
- a. SQL
 - b. Oracle
 - c. MySQL
 - d. MongoDB

Consider the following figures for questions 9 through 12

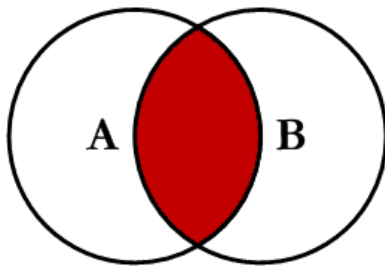


Figure 9.1

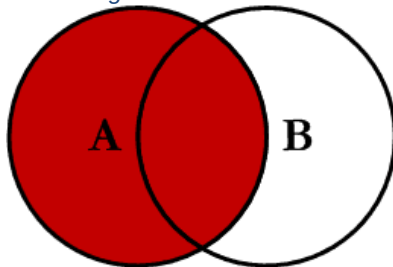


Figure 9.2

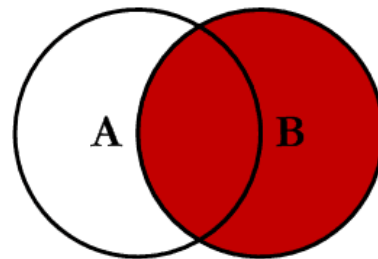


Figure 9.3

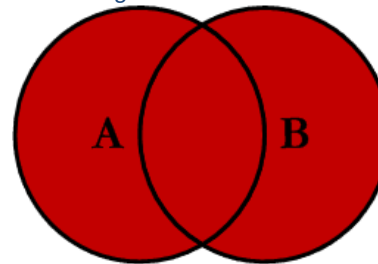


Figure 9.4

9. Which of the above figures refer to an OUTER JOIN?
- a. Figure 9.1
 - b. Figure 9.2
 - c. Figure 9.3
 - d. Figure 9.4
10. Which of the above figures refer to an RIGHT JOIN?
- a. Figure 9.1
 - b. Figure 9.2
 - c. Figure 9.3
 - d. Figure 9.4
11. Which of the above figures refer to an LEFT JOIN?
- a. Figure 9.1
 - b. Figure 9.2
 - c. Figure 9.3
 - d. Figure 9.4

12. Which of the above figures refer to an INNER JOIN?
- a. Figure 9.1
 - b. Figure 9.2
 - c. Figure 9.3
 - d. Figure 9.4

Refer to the following code to answer questions 13 through 17.

- a. SELECT ProductName, Price, InvQuantity
 - b. FROM Inventory.Products
 - c. WHERE InventoryProductSubCategoryID IN
 - d. (SELECT InventoryProductSubCategoryID
 - e. FROM Inventory.ProductSubcategory
 - f. WHERE ProductName = 'Computers'
13. The step executed in the above inner query (line d to f) finds the *ProductName* from *ProductSubCategory* table
- a. True
 - b. False
14. The syntax "Inventory" on line b refers to
- a. Table Name
 - b. Schema Name
 - c. Database Name
 - d. None of the above
15. The above SQL statement is evaluated in how many steps?
- a. 3
 - b. 1
 - c. 2
 - d. 4
16. The step within the outer query evaluated in the above SQL statement (line a to c) finds the product subcategory from the field *ProductName* selected in the inner query.
- a. True
 - b. False
17. What key definition in the *Products* and *ProductSubCategory* tables from the SQL statement above is most probable?
- a. The field *Name* is a primary key in the *Product* table
 - b. The field *ProductSubcategoryID* is the primary key in the *Product* table
 - c. The field *Name* is a foreign key in the *ProductSubcategory* table
 - d. The field *ProductSubcategoryID* is a foreign key in the *Products* table

Refer to the following data set below for questions 18 and 19

{10, 20, 20, 35, 70, 70, 90, 100, 115, 115}

18. What is the result of running a NTILE on the set, if we were to make 3 buckets on the above dataset?
- a. Bucket 1: 115, 115, 100
Bucket 2: 90, 70, 70
Bucket 3: 35, 20, 20, 10
 - b. Bucket 1: 10, 20, 20, 35
Bucket 2: 70, 70, 90
Bucket 3: 100, 115, 115
 - c. Bucket 1: 115, 115
Bucket 2: 100, 90
Bucket 3: 70, 70, 35, 20, 20, 10
 - d. None of the above
19. What is the result of running a DENSE_RANK on the set?
- a. {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
 - b. {1, 2, 2, 4, 5, 5, 7, 8, 9, 9}
 - c. {1, 2, 2, 3, 4, 4, 5, 6, 7, 7}
 - d. None of the above
20. What is a set of technologies that is used for copying and distributing data and database objects from one database to another?
- a. Replication
 - b. Database Logging
 - c. Database Mirroring
 - d. None of the above
21. What issues does replication help overcome within an SQL Server?
- a. Availability problems
 - b. Maintenance problems
 - c. Performance problems
 - d. All of the above
22. Which of the following is used in order to make data available to be replicated from one server to another?
- a. Subscription
 - b. Publisher
 - c. Article
 - d. Distributor

23. Which of the following is *not* a typical use scenario of transactional replication?
- a. Integrating data from multiple sites
 - b. Data warehousing and reporting
 - c. Mobile applications or distributed server applications that have possible data conflicts.
 - d. Server-to-server scenarios that require high throughput, including: improving scalability and availability
24. Which of the following are types of replications?
- a. Transactional
 - b. Snapshot
 - c. Merge
 - d. All of the above
25. Which of the following are types of replication agents?
- a. Snapshot Agent
 - b. Merge Agent
 - c. Queue Reader Agent
 - d. All of the above
26. Which of the following replication agents are used to move transactions that are ready to be replicated from the transaction log within the Publisher to the distributor database?
- a. Log Reader Agent
 - b. Queue Reader Agent
 - c. Snapshot Agent
 - d. Distribution Agent
27. Which of the following replication agents are used to move transactions that are currently held in the distributor database and pushes them to the subscribers?
- a. Merge Agent
 - b. Distribution Agent
 - c. Snapshot Agent
 - d. Log Reader Agent
28. Replication can occur no matter if the database recovery model is simple or full.
- a. True
 - b. False
29. The following clause(s) can contain search JOINS:
- a. HAVING
 - b. FROM
 - c. WHERE
 - d. All of the above

30. Subqueries *cannot* be nested inside which of the following statements?
- a. CREATE
 - b. DELETE
 - c. SELECT
 - d. UPDATE
31. Which of the following is *true* about the result of a sub-query?
- a. The result of a sub-query is always NULL
 - b. The result of a sub-query is generally ignored when executed
 - c. The result of a sub-query is used by the main query
 - d. The result of a sub-query doesn't give a result; it is just helpful in speeding up the main query execution.
32. Which of the following is a method for writing a sub-query in a main query?
- a. By using WHERE clause
 - b. By writing a SELECT statement embedded in the clause of another SELECT statement
 - c. By using the GROUP BY clause
 - d. By using JOINS
33. Which of the following operators *cannot* be used in a sub-query?
- a. < >
 - b. AND
 - c. >
 - d. <
34. What is *true* about the ALL operator used for sub-queries?
- a. Returns rows only if all the values match in a list/sub-query
 - b. Returns rows that match only some values in a list/sub-query
 - c. Returns rows that match all the values in a list/sub-query
 - d. All of the above
35. Which of the following statements *cannot* be parent statements for a sub-query?
- a. UPDATE
 - b. DELETE
 - c. SELECT
 - d. GROUP BY
36. What standard did SQL become a part of in 1986 and 1987?
- a. American National Standards Institute
 - b. International Organization for Standardization
 - c. Both a and b
 - d. None of the above

37. Which of the following is the bitwise operator for NOT?
- $\mid =$
 - $\sim$
 - $\wedge =$
 - $\& =$
38. A deadlock exists in the system if and only if the wait for graph:
- Has a cycle in it
 - Is a tree
 - Has a path from first node to last node
 - None of the above
39. The number of entities to which another entity can be associated via a relationship set is expressed as:
- Attributes
 - Schema
 - Entity
 - Cardinality
40. SQL view is said to be updatable (that is, inserts, updates or deletes can be applied on the view) if which of the following conditions are satisfied by the query defining the view?
- The query does not have a group by or having clause
 - The from clause has only one database relation
 - The select clause contains only attribute names of the relation, and does not have any expressions, aggregates, or distinct specification
 - All of the above
41. Which of the following is *true* regarding views?
- A user who creates a view receives all privileges on that view
 - If a user creates a view on which no authorization can be granted, the system will allow the view creation request
 - The user who creates a view cannot be given update authorization on a view without having update authorization on the relations used to define the view
 - The user who creates a view cannot be given update authorization on a view without having update authorization on the relations used to define the view
42. Domain constraints, functional dependency and referential integrity are special forms of?
- Referential constraint
 - Foreign key
 - Assertion
 - Primary key

43. Which of the following can be addressed by enforcing a referential integrity constraint?
- a. Information on the customer must be known before anything can be sold to that customer
 - b. When entering an order quantity, the user must input a number and not some text (i.e., 12 rather than 'a dozen')
 - c. Certain fields are required (such as the email address, or phone number) before the record is accepted
 - d. All phone numbers must include the area code

44. Which statement do you use to assign an initial value to a variable
- a. PARAMETER
 - b. CALL
 - c. ASSIGN
 - d. DECLARE

45. Which of the following *best* describes what would be returned by a query with the following criteria?

WHERE City LIKE '%nd';

- a. All cities that have nd in them
 - b. All cities that have nd and a percent sign in it
 - c. All cities that have nd at the end
 - d. None of the above
46. IF-THEN-ELSE logical processing *can* be accomplished within an SQL statement
- a. True
 - b. False
47. What of the following is *not* true of a Differential Backup?
- a. Ability to allow retention of several versions of the same files
 - b. Files and folders are backed up to one backup set
 - c. Full backup and last differential backup needed for restore
 - d. Less storage space needed
48. There are differences between stored procedures and functions. Which of the following is *not* a difference?
- a. Stored procedures can be utilized within a SELECT statement while functions can't be embedded in a SELECT statement
 - b. Stored procedures can use SELECT, INSERT, UPDATE, DELETE while functions only use SELECT
 - c. Stored Procedures have input/output parameters while functions only have input parameters
 - d. Functions return a value while stored procedures aren't require to return a value.

49. Which of the following is *true* about stored procedures?
- a. A stored procedure uses the DECLARE keyword in the procedure body to declare formal parameters.
 - b. A stored procedure is named PL/SQL block with at least one parameter declaration in the procedure specification.
 - c. A stored procedure uses the DELCLARE keyword in the procedure specification to declare formal parameters.
 - d. A stored procedure must have at least one executable statement in the procedure body.
50. Consider the following Transact-SQL statement fragment from a stored procedure. What happens if an error occurs during the INSERT statement?

```
INSERT INTO Temp (Col1)
VALUES @TempVal
PRINT @TempVal
IF @@error <> 0
```

```
RETURN 1
ELSE
```

```
RETURN 0
...MORE CODE BELOW...
```

- a. The stored procedure exits with no return value
- b. The stored procedure continues to execute more code below
- c. The stored procedure exits with a return value of 1
- d. The stored procedure exits with a return value of 0